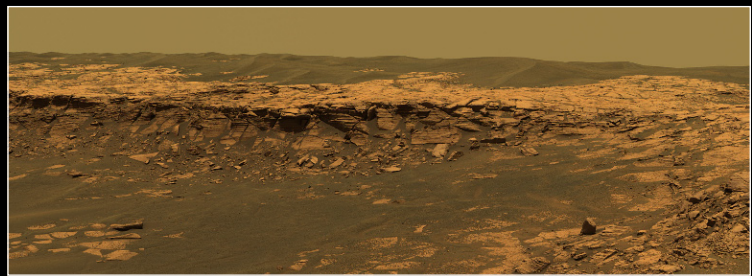


THE PLANETS

The eight solar system planets all fall into one of two categories: the rocky planets and the gas giants. The rocky planets are the four closest to the Sun—Mercury, Venus, Earth, and Mars. They were formed from rocky and metallic material near the newly born Sun. Each consists of a rocky mantle and crust, surrounding a metal core, yet their surfaces differ widely. Mercury and Venus are hot, lifeless, dry worlds, but while the former is covered in craters, the latter is shrouded in

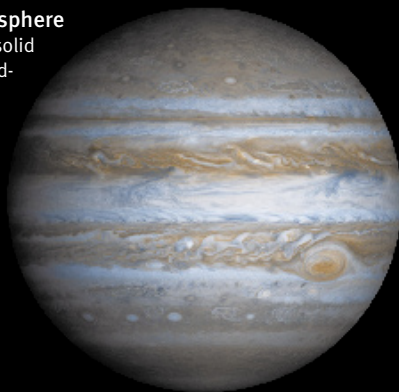


Rocky landscape

This panoramic view of Mars was produced from 28 images taken in February 2006 by the rover *Opportunity* during its exploration of the planet. The view is of the western edge of Erebus Crater. Layered rocks can be seen in the 3-ft- (1-m-) thick crater wall.

Gaseous atmosphere

The gas giants do not have solid surfaces; what we see is the cloud-top layer of their atmospheres. Jupiter's visible surface shows bands of swirling gas—and spots, which are giant storms.



PLANET TYPES

ROCK	GAS
Mercury	Jupiter
Venus	Saturn
Earth	Uranus
Mars	Neptune

more than 90 percent of asteroids orbit in the main belt between Mars and Jupiter

Saturn is the second-largest planet and is nearly ten times as far from the Sun as Earth is

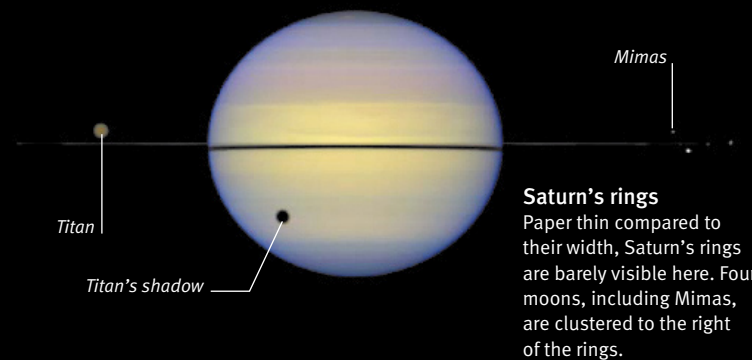
the smallest gas giant and most distant planet, Neptune is 30 times farther from the Sun than Earth is

a thick atmosphere that hides a volcanic landscape. By further contrast, Earth is wet and teems with life, and the more distant, colder Mars is a red world of frozen desert.

The gas giants Jupiter, Saturn, Uranus, and Neptune are the four largest planets. They formed from rock, metal, gas, and ice in the cooler outer regions of the disk of material surrounding the young Sun. Each has a rock-rich core surrounded by an unfathomably deep and thick atmosphere. All four have a ring system encircling them and a large number of orbiting moons.

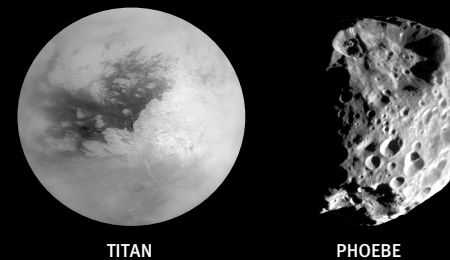
MOONS AND RINGS

At present, 162 planetary moons are known to exist, but more are likely to be found as observing techniques improve. The majority belong to the gas giants; only Mercury and Venus are moonless. The moons are made of either rock or a mix of rock and ice, and they orbit around their planets like mini solar systems. The largest is Jupiter's moon Ganymede, which is bigger than Mercury. The smallest are hill-sized irregular lumps. All four giant planets also have ring systems comprising pieces of dirty ice. The most extensive of these is the one around Saturn.



Contrasting moons

Titan is the second-largest moon in the solar system, and the only one with a substantial atmosphere. A nitrogen-rich smoggy haze envelops the moon, but here it has been removed to show the moon's surface. Phoebe, another of Saturn's moons, is more typical: it is 143 miles (230 km) long, potato-shaped, and covered in craters.



TITAN

PHOEBE

MINOR MEMBERS

There are more than a trillion smaller bodies in the solar system. The majority are comets and asteroids, followed by thousands of Kuiper belt objects, and finally the dwarf planets: Eris and Pluto, two icy rock bodies beyond Neptune; and rocky Ceres in the main belt of asteroids. Over a billion asteroids are thought to exist. More than 200,000 are in the Main Belt between Mars and Jupiter—these are the remnants of a failed process of planet formation. Most of the asteroids are irregularly shaped lumps of rock. Comets are huge, dirty snowballs within the Oort cloud. When one leaves the cloud and travels in toward the Sun, it becomes large enough and bright enough to be seen.



Comet Hale-Bopp

Hale-Bopp was seen in 1997 as it traveled close to the Sun. Its large head and its tails of gas (blue) and dust (white) made it one of the brightest 20th-century comets.



Asteroid Eros

Orbiting outside the main belt between Mars and Earth, the potato-shaped Eros is 19 miles (31 km) long. Its surface is covered in craters where other bodies have struck it.

WHAT'S IN THE SOLAR SYSTEM?